SSFNet: Spectral–Spatial Fusion Network for Hyperspectral Remote Sensing Scene Classification

Zhen Li, Puhong Duan, *Member, IEEE*, Jialin Zheng, Zhuojun Xie, Xudong Kang, *Senior Member, IEEE*, Jianwei Yin, *Member, IEEE*, and Shutao Li, *Fellow, IEEE*

***Abstract*— Remote sensing scene classification (RSSC) plays a vital role in a variety of applications and has attracted much more attention. In recent years, much progress has been made to release diverse datasets or develop all kinds of techniques for scene classification of multispectral remote sensing images. Nevertheless, very few studies have focused on hyperspectral image scene classification. Moreover, the existing scene classification approaches fail to fully employ the rich spectral information of the input images, which cannot achieve satisfactory performance for hyperspectral images. To alleviate these issues, this work proposes a spectral–spatial fusion network (SSFNet) for hyperspectral RSSC (HRSSC). First, a multiscale regional growth search (MSRGS) method is designed to extract salient object regions from the hyperspectral remote sensing scene. Then, a three-stream network architecture is proposed to extract the global spatial, local spatial, and spectral features, respectively. Finally, the fully connected layer is performed on the extracted features to obtain a class score followed by a decision fusion scheme to generate the final classification result. To evaluate the effectiveness of the proposed SSFNet, we created a publicly available benchmark for the HRSSC dataset, which contains 1445 hyperspectral images, covering 11 scene classes. Experiments on the HRSSC database claim that the proposed SSFNet can attain superior classification performance with respect to other state-of-the-art scene classification techniques. The code of the proposed SSFNet will be available at https://github.com/PuhongDuan/SSFNet.**

***Index Terms*— Deep learning, hyperspectral data, scene classification, spectral–spatial fusion.**

Received 5 December 2024; revised 16 February 2025; accepted 24 February 2025. Date of publication 7 March 2025; date of current version 27 March 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFA0715203; in part by the National Natural Science Foundation of China under Grant 62221002, Grant 62201207, and Grant 62101183; in part by the Natural Science Foundation of Hunan Province under Grant 2023JJ40163; and in part by the Science and Technology Innovation Program of Hunan Province under Grant 2023RC3124 and Grant 2024RC1030. *(Corresponding author: Puhong Duan.)*

Zhen Li is with the College of Computer Science and Technology (CCST), Zhejiang University, Hangzhou 310027, China, and also with the Institute of Remote Sensing Satellite, China Academy of Space Technology (CAST), Beijing 100094, China (e-mail: lizhen_cast@zju.edu.cn).

Puhong Duan, Zhuojun Xie, Xudong Kang, and Shutao Li are with the School of Robotics, Hunan University, Changsha 410082, China (e-mail: puhong_duan@hnu.edu.cn; xzj22@hnu.edu.cn; xudong_kang@163.com; shutao_li@hnu.edu.cn).

Jialin Zheng is with Pinghu Laboratory, Pinghu, Zhejiang 314200, China (e-mail: zjialin@hnu.edu.cn).

Jianwei Yin is with the College of Computer Science and Technology, Hangzhou 310027, China, also with the School of Software Technology, Zhejiang University, Ningbo 315048, China, and also with China Institute for New Urbanization Studies, Huzhou 313000, China (e-mail: zjuyjw@cs.zju.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3549075

I. Introduction

With the advancement of remote sensing imaging technology, numerous semantic-level tasks have been involved in recent years. These tasks encompass a great deal of applications, including but not limited to image classification [1], [2], [3], object detection [4], [5], [6], [7], image captioning [8], [9], [10], and scene segmentation [11], [12]. Among these tasks, scene classification of remote sensing image is a fundamental mission, with the aim of assigning a unique high-level semantic category to each scenario [13] and has become a hotspot topic, which plays an important role in city mapping, scene positioning, and image understanding. Over the past few years, many scholars have conducted extensive researches on scene classification tasks in remote sensing field and proposed a series of classification techniques, which can be roughly grouped into three classes [14]: manual feature-based approaches, unsupervised feature-based learning approaches, and deep learning approaches.

In the early stage, scholars mainly select image feature extraction techniques for scene classification, which mainly relies on expert knowledge to select color, texture, scale, and shape features to identify and classify remote sensing images. The classical methods include local binary pattern (LBP), color histogram, scale-invariant feature transformation (SIFT), directional gradient histogram (HOG), etc. Accordingly, the classification performance of these methods heavily relies on manually selected features. Unsupervised feature learning methods autonomously learn mapping functions for feature encoding. They take either handcrafted features or original pixels as input and generate a series of high-order features. By automatically learning representative features from unlabeled data rather than manually selected features, the extracted features have a strong representation ability. The popular unsupervised techniques have autoencoder (AE), sparse coding, and principal component analysis (PCA). These methods are capable of achieving better classification performance compared to traditional handcrafted feature-based approaches. Nevertheless, due to the lack of class labels, this type of method may not ensure optimal discriminative ability among different classes.

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:56:11 UTC from IEEE Xplore. Restrictions apply.

Different from traditional feature extraction methods, deep learning methods can deeply mine advanced semantic features in remote sensing images and therefore exhibit satisfactory performance in feature extraction and classification tasks [15], [16]. With the rapid advancement of computer science and remote sensing imaging technology, an increasing number of researchers are applying deep learning networks for remote sensing scene classification (RSSC) tasks. For example, Minetto et al. [17] proposed an ensemble framework based on convolutional neural networks (CNNs) for the RSSC task. Wang et al. [18] developed a novel dual branch framework for RSSC, which uses a key region search strategy. Xu et al. [19] introduced a double branch network with unshared parameters, which was designed to mine the features from both of the two input images. Cao et al. [20] applied the self-attention network to extract and fuse the features that polymerized from multiple layers and enhanced the importance of intricate objects within remote sensing scenes. Wang et al. [21] developed a novel recursive attention structure method for RSSC. Tang et al. [22] proposed an attention-consistent deep model (ACNet) that can effectively extract global features, and the model is able to effectively align features of important areas well. Chen et al. [23] introduced a new framework named global context spatial attention (GCSA) for RSSC. This approach effectively extracts multiscale global scene features. He et al. [24] proposed a skip-connected covariance (SCCov) deep model to utilize the second-order attributes of high-resolution images and fully integrated the information of multiresolution feature maps. Fang et al. [25] designed a joint learning module for remote sensing scene feature extraction, which comprehensively utilized spatial-frequency domain discriminant features.

These methods mentioned above can boost the classification accuracy for RSSC. However, both of them are designed for three-channel remote sensing data, such as the UC Merced Land-Use dataset [26], WHU-RS19 dataset [27], RSSCN7 dataset [28], AID dataset [29], and NWPU-RESISC45 dataset [14]. Only spatial information is used to identify different kinds of scenes. When the visual features of two remote sensing images are very similar, these methods are prone to yield misclassification and confusing results. Different from multispectral remote sensing images, hyperspectral images not only comprise spatial information but also include rich spectral information of the imaging scene. This characteristic makes hyperspectral images more effective for the RSSC. Nevertheless, very few studies focus on the scene classification of hyperspectral images. Owing to the abundant spectral information of HSIs, the existing scene classification approaches for multispectral images cannot yield satisfactory classification performance when they are directly applied to HSIs. The reason is that most of them cannot fully utilize the rich spectral information in HSIs for scene classification.

To solve these issues, we propose a spectral–spatial fusion network (SSFNet) for scene classification of hyperspectral remote sensing images, which aims to assign hyperspectral images to a predefined class. The proposed method is a three-branch network architecture that consists of the global branch, the local spatial branch, and the local spectral branch, respectively. Specifically, a multiscale regional growth search (MSRGS) method is first proposed to extract the object region from HSI. Then, the local spectral and spatial features are calculated by using a deep feature extraction network on the local object region, and the global spatial feature is extracted by performing the deep CNN on the RGB image. Finally, the obtained local spectral and spatial features are integrated together followed by the fully connected layer, and then, the class scores obtained by the global and local branches are merged to yield the final classification result. To further assess the effectiveness of the proposed SSFNet, a hyperspectral remote sensing scene classification (HRSSC) dataset is constructed, which consists of 1445 hyperspectral images, divided into 11 different scenes. Experiments on HRSSC dataset imply that our proposed SSFNet framework obtains outstanding classification performance compared to other state-of-the-art RSSC approaches. The key contributions of this article can be summarized as follows.

1) A novel and publicly available benchmark dataset for hyperspectral RSSC is constructed, which consists of 11 classes with 1445 hyperspectral images.[1] To our knowledge, the HRSSC dataset is the largest scale on the total number of scene categories and the amount of hyperspectral data. The HRSSC can be used to assess the effectiveness of novel proposed methods, which will facilitate the development for future research.
2) To take full advantage of the rich spectral and spatial information in HSIs, an SSFNet architecture is proposed for HRSSC task, in which a three-branch network architecture is constructed to extract the global spatial feature, the local spatial feature, and the local spectral feature, respectively.
3) To determine the location of key area in hyperspectral remote sensing images, an MSRGS method is proposed to accurately calculate the coordinates of important region. This strategy is beneficial to extract the local important features while reducing the influence of useless information in the imaging scene.

The main structure of this article is outlined as follows. Section II provides a concise review of relevant literature. Section III provides a detailed presentation of the proposed SSFNet. The experimental parameter settings and results are presented in Section IV. Lastly, Section V offers a summary of the conclusions and discussions.

II. Related Work

In this section, we will briefly introduce the related studies for RSSC. According to the feature extraction manners, RSSC can be approximately divided into three aspects: manual feature extraction [30], [31], unsupervised feature learning [32], [33], and deep feature learning [34], [35].

The manual feature-based approaches were the earliest applied in the RSSC task. The core idea of this type of method is to design different feature descriptors. For example, Huang et al. [36] employed different scales of LBPs for RSSC, in which the kernel Fisher encodes were utilized to improve the discrimination of image features. Li et al. [37]

[1] https://pan.baidu.com/s/1H-iRye0-HkxgfTlAHDWe1w (code: HSCD).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:56:11 UTC from IEEE Xplore. Restrictions apply.

introduced a novel approach called tensor gradient SIFT description algorithm to mine rich spatial information, improving the classification accuracy. To improve the classification performance, multiple features are jointly employed for scene classification. For instance, Luo et al. [38] fused five types of image features, that is, radiometric features, texture features (including the Gaussian wavelet features, the Gabor features, and the Gray Level Co-occurrence Matrix features), and the shape features, to represent remote sensing images, which was applied for RSSC with different resolutions of data. Zou et al. [39] put forward a collaborative representation approach of RSSC, in which the multiscale completed LBPs were exploited to extract the global features of the input, and the local image features were extracted by the bag of visual words. Although these methods can achieve the task of RSSC, it is hard to select the optimal features for real-world remote sensing images [40].

The unsupervised learning approaches for RSSC [32], [41], [42], [43] focus on automatically learning mapping functions for feature encoding. These methods consider a series of handcrafted features or original images as input in the learning process. For instance, Chan et al. [32] developed a learnable PCA to extract robust invariant features for RSSC. Zheng et al. [41] built a large data annotation framework that combined spatial constraints to more accurately annotate image categories for the RSSC task. Zhang et al. [42] proposed an AE method with sparse to characterize the complex satellite image scene. Bahmanyar et al. [43] used an unsupervised feature learning method to generate visual word bag models, which were applied for RSSC. These unsupervised feature learning-based methods can achieve excellent results in RSSC than the handcrafted feature-based techniques. However, owing to being short of corresponding supervised label information, they cannot generate the optimal discrimination capability among different land covers.

In recent years, the CNN [14], [21], [35], [44], [45], [46], [47] models in the deep learning network have shown huge potential for the RSSC task due to their strong feature characterization ability. A considerable number of deep learning-based techniques have been applied for RSSC. For example, Zhao et al. [48] designed a multigranularity classification network for RSSC with a multilevel feature ensemble, which can ensemble multiple different types of information together. Zhang et al. [49] developed an attribute-guided CNN model for scene classification; the additional attribute information of which was utilized as constraint. In addition, some attention mechanisms have been also used for RSSC [50], [51], [52]. These RSSC approaches usually need extensive training data and high computing resource. To alleviate this issue, some few-shot learning frameworks have been investigated. For example, Zhang et al. [53] designed a multibranch feature extraction networks-based few-shot RSSC framework, in which an online sample generation technique was adopted at the feature space. Xiong et al. [54] combined FixMatch and weighted nuclear-norm regularization to solve small sample issue, in which two weighting schemes were used to boost the prediction discriminability. Li et al. [55] proposed a heterogeneous remote sensing data fusion method based on copula-guided neural network, providing enhanced data features with explainability and superior performance for change detection tasks. Li et al. [56] applied self-supervised learning for RSSC task with limited samples, which can effectively improve the feature extraction ability of the classification network by utilizing the unlabeled data in the training phase. Although these deep feature learning-based RSSC approaches have made significant progress, they are constructed for multispectral images with limited spectral bands, which do not utilize the spectral information from source images. Therefore, when these approaches are directly applied for HRSSC, they tend to yield unsatisfactory classification performance.

III. Proposed Method

In this section, the detailed steps of our SSFNet method are described. First, the network architecture is briefly introduced. Then, the MSRGS method is presented. Finally, the feature extraction method is introduced.

A. Network Architecture

The flowchart of the proposed SSFNet is depicted in Fig. 1, which mainly contains object region detection, spectral and spatial feature extraction, and decision fusion. First, an MSRGS method is designed to extract the important object region and reduce the interference of useless details.

$$\mathbf{M} = \texttt{MSRGS}(\mathbf{I}) \tag{1}$$

where $\mathbf{M} \in \mathbb{R}^{N_s \times H \times W}$ is the local hyperspectral image that mainly reflects spectral–spatial information in the salient object region. $\mathbf{I} \in \mathbb{R}^{N \times H \times W}$ denotes the input hyperspectral image. MSRGS represents an MSRGS operation.

Then, a three-branch network framework is designed to extract the global spatial, the local spatial, and the local spectral features. Here, the local spectral feature is acquired by using average pooling on local hyperspectral image. The local spatial feature is calculated by using the backbone on the local RGB image, and the global spatial feature is estimated by performing the backbone on the RGB image.

$$\mathbf{S}_l = \text{FC}(\text{AvgPool}(\mathbf{M})) \tag{2}$$

$$\mathbf{F}_l = \text{AvgPool}(\mathcal{F}(\mathbf{M}_{\text{RGB}})) \tag{3}$$

$$\mathbf{F}_g = \text{AvgPool}(\mathcal{F}(\mathbf{I}_{\text{RGB}})) \tag{4}$$

where AvgPool is the average pooling operation, and $\mathcal{F}$ is the feature extraction operation. $\mathbf{S}_l$ represents the average spectral feature. $\mathbf{F}_l$ and $\mathbf{F}_g$ indicate the local and global features, respectively. FC is the fully connected layer that is utilized to increase the dimension of feature vectors.

Finally, the local spatial and spectral features are first merged followed by the FC layer to yield the class score, and then, the class scores of the local and global features are integrated to produce the final class score.

$$\mathbf{C} = \text{FC}\left(\mathbf{F}_g\right) + \text{FC}(\mathbf{S}_l + \mathbf{F}_l) \tag{5}$$

where $\mathbf{C}$ is the final classification score.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:56:11 UTC from IEEE Xplore. Restrictions apply.

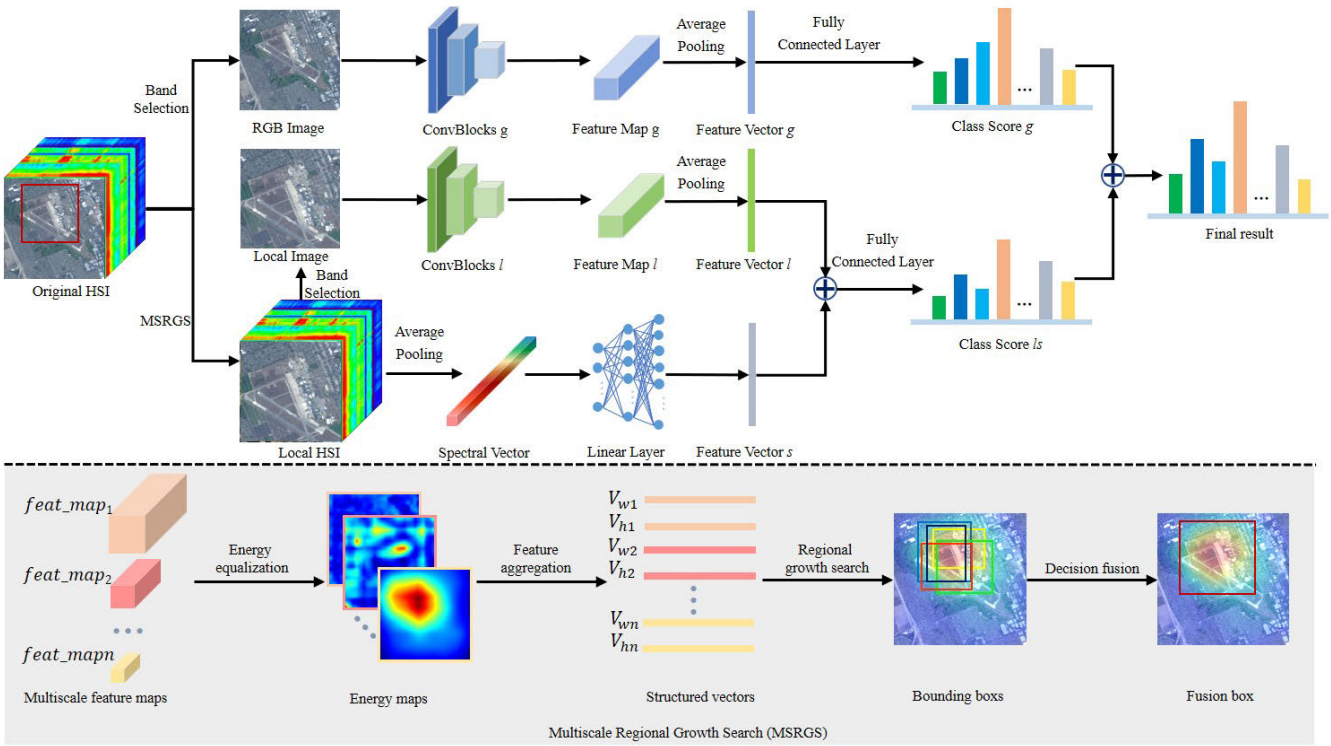


Fig. 1. Overall framework of the proposed SSFNet.

B. MSRGS

Since the imaging scene not only contains object information but also records a large amount of background interference information. This background information may have a negative effect on scene classification especially for spectral feature extraction. A simple solution is to extract the object region from the original HSIs. In this work, an MSRGS method is proposed to extract the local key region. Specifically, we first perform averaging operation along with the channel dimension on the obtained features of each layer obtained by the feature extractor.

$$\mathbf{M}_{F_i} = \mathrm{Avg}(\mathbf{F}_i) \tag{6}$$

where Avg denotes averaging operation. $\mathbf{F}_i$ is the output features of ith layer. $\mathbf{M}_{F_i}$ is the fused feature. It should be noted that each fused feature is scaled into [0, 1] so as to remove the interference of negative elements.

$$\widehat{\mathbf{M}_{\mathbf{F}}}(i) = \frac{\mathbf{M}_{F_i} - \min(\mathbf{M}_{F_i})}{\max(\mathbf{M}_{F_i}) - \min(\mathbf{M}_{F_i})} \tag{7}$$

where $\max(\mathbf{M_F})$ is the maximum value in $\mathbf{M}_{F_i}$, and $\min(\mathbf{M_F})$ is the minimum value in $\mathbf{M}_{F_i}$. $\widehat{\mathbf{M}_{\mathbf{F}}}(i)$ is the normalized feature map.

Then, to locate object region quickly and efficiently, we convert the 2-D energy map to a 1-D vector by using feature aggregation in different dimensions.

$$\begin{cases} \mathbf{V}_w = \sum_{i=1}^{H} \widehat{\mathbf{M}_{\mathbf{F}}}(i, W) \\ \mathbf{V}_h = \sum_{i=1}^{W} \widehat{\mathbf{M}_{\mathbf{F}}}(H, i) \end{cases} \tag{8}$$

where $\mathbf{V}_w$ and $\mathbf{V}_h$ are the 1-D energy vector of $\widehat{\mathbf{M}_{\mathbf{F}}}$. H and W represent the total number of rows and columns in the horizontal and vertical directions, respectively. This operation offers two benefits. Firstly, it significantly enhances search efficiency by transforming the key area search into two separate searches along 1-D energy vectors. Secondly, it enables decoupling in the 2-D space through independent searches along both spatial width and high directions.

Next, we develop a regional growth search method on the 1-D energy vector $\mathbf{V}_w$ and $\mathbf{V}_h$ to obtain key region coordinates, that is, $[x1, x2]$ and $[y1, y2]$. Algorithm 1 presents the overall process of the regional growth search method. Taking $\mathbf{V}_w$ as an example, the specific process is shown as below.

Step A: (Initializing the Seed Point): We calculate the index of maximum value in $\mathbf{V}_w$ as the seed point.

$$x_1 = \underset{i}{\mathrm{argmax}}(\mathbf{V}_w). \tag{9}$$

Step B: (Updating the Search Region): We estimate the search region along with neighboring pixels in the height direction.

$$E_{[x_1:x_2]} = \sum_{i=x1}^{x2} \mathbf{V}_w(i) \tag{10}$$

$$E_S = \sum_{i=1}^{H} \mathbf{V}_w(i) \tag{11}$$

where E_S indicates the sum of the entire vector values, and $E_{[x1,x2]}$ denotes the region energy along with the spatial height from x_1 to x_2. If $E_{[x_1:x_2=x_1+1]}$ is higher than $E_{[x_1:x_2=x_1-1]}$, the search region is extended along the right side in a pixel-by-pixel

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:56:11 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Regional Growth Search

Input: 1-D energy vector $\mathbf{V}_w \in \mathbb{R}^H$, $T \in (0, 1]$;
Output: The coordinates of key region: $[x_1, x_2]$;

$x_1 = \text{argmax}(\mathbf{V}_w)$
$x_2 = x_1 + 1$ or $x_2 = x_1 - 1$
if $E_{[x1:x1+1]} > E_{[x1-1:x1]}$ **then**
  $x_2 = x_1 + 1$
else
  $x_2 = x_1$
  $x_1 = x_1 - 1$
end if
$T_S = E_{[x_1:x_2]} / E_S$
while $T_S <$ T **do**
  if $E_{[x_1-1:x_2]} > E_{[x_1:x_2+1]}$ **then**
    $x_1 = x_1 - 1$
  else
    $x_2 = x_2 + 1$
  end if
  $T_S = E_{[x_1:x_2]} / E_S$
end while
$x_1 = \frac{x_1}{H} \times 100\%$
$x_2 = \frac{x_2}{H} \times 100\%$
return $[x_1, x_2]$

manner. Otherwise, the search region is on the left side. When $E_{[x1:x2]} > T \times E_S$, the region will be considered the local object area in the height direction, where $T \in (0, 1]$ is a threshold that controls the size of the key area.

Step C: (Scaling the Boundary): Since the obtained boundary values are varying from 0 to H, it is necessary to scale them to [0, 1].

$$x_1 = \frac{x_1}{H} \times 100\%$$
$$x_2 = \frac{x_2}{H} \times 100\%. \tag{12}$$

Based on a series of operations mentioned above, the height boundary of the local object region in HSI, that is, $[x_1, x_2]$, can be obtained. Similarly, the width boundary $[y_1, y_2]$ can be also estimated by performing the same operations. Therefore, the object boundary $[x_1, x_2, y_1, y_2]$ in each layer feature can be obtained. In this article, to take full advantage of the multiscale feature information, the object boundary of each layer feature is calculated. Finally, the entire boundary in HSI is obtained by fusing multiple object boundaries.

$$\begin{aligned}[x_s, y_s] &= \left[\min\{x_{s_i}\}, \min\{y_{s_i}\}\right] \\ [x_e, y_e] &= \left[\max\{x_{e_i}\}, \max\{y_{e_i}\}\right]\end{aligned} \tag{13}$$

where x_{s_i} is the upper left corner coordinate point of ith feature map in the horizontal direction. x_{e_i} is the bottom right corner coordinate point of ith feature map in the horizontal direction. y_{s_i} denotes the upper left corner coordinate point of ith feature map in the vertical direction. y_{e_i} denotes the ending point of ith feature map in the vertical direction. Through these operations, the final boundary $[x_s, x_e, y_s, y_e]$ is obtained, which is utilized to guide the local object area extraction.

C. Feature Extraction

To extract image features of the input data, two representative feature extraction architectures, that is, CNN and transformer, are considered in the SSFNet framework. The details of which two network architectures are described as follows.

1) CNN: Generally, when using the CNN model for image classification tasks, it has three components: convolutional layer, pooling layer, and fully connected layer. The convolutional layer is mainly responsible for extracting image features, including low-level color and texture features, as well as advanced semantic features. Each convolution layer typically consists of a convolution kernel, a nonlinear activation function, and batch normalization [57]. The pooling layer reduces network computing and extracting the most significant features by reducing the dimensionality and quantity of feature maps, extracting the most significant features. And also the pooling layer is employed to connect the convolutional layers and the FC layer while the FC layer is to convert the input features into class scores. Since different CNNs have different network architectures, such as residual network (ResNet), AlexNet, and GoogleNet, it is difficult to depict their contents one by one. Here, we uniformly use ConvBlocks to represent the feature extraction process. Assume $\mathbf{I} \in \mathbb{R}^{N \times H \times W}$ be the input image, and the multilayer feature maps $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$($C$, H, and W represent the amount of channels, height, and width, respectively) can be expressed as follows:

$$\mathbf{F} = \text{ConvBlocks}(\mathbf{I}) \tag{14}$$

where ConvBlocks denotes a deep convolution feature extraction module.

2) Transformer: Different from CNNs, transformer is a self-attention mechanism-based deep network, which has been widely applied in image and text processing tasks. In this work, a mix transformer (MIT) network [58] is adopted as the backbone, in which each encoder has three modules, that is, multihead self-attention (MSA), mix feed-forward network (MFFN), and overlapped patch merging (OPM). The most important part of the encoder is the self-attention mechanism, which is estimated as follows:

$$\begin{aligned}\widehat{\mathbf{K}} &= \text{Reshape}\left(\frac{N}{R}, C \cdot R\right)(\mathbf{K}) \\ \mathbf{K} &= \text{Linear}(C \cdot R, C)(\widehat{\mathbf{K}}) \\ \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) &= \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_{\text{head}}}}\right)\mathbf{V}\end{aligned} \tag{15}$$

where R is a reduction ratio. Reshape $((N/R), C \cdot R)(\mathbf{K})$ refers to reshape K to the one with dimension of $((N/R), C \cdot R)(\mathbf{K})$. The spatial dimension of $\mathbf{Q}$ is $N \times C$, and the spatial dimension of $\mathbf{K}$, $\mathbf{V}$ is $(N/R) \times C$. Algorithm 2 depicts the overall pseudocode of the proposed SSFNet.

IV. Experiments

In this section, we designed several experiments to verify the effectiveness of our SSFNet classification framework. First, the experimental setup is introduced. Then, we conducted ablation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:56:11 UTC from IEEE Xplore. Restrictions apply.

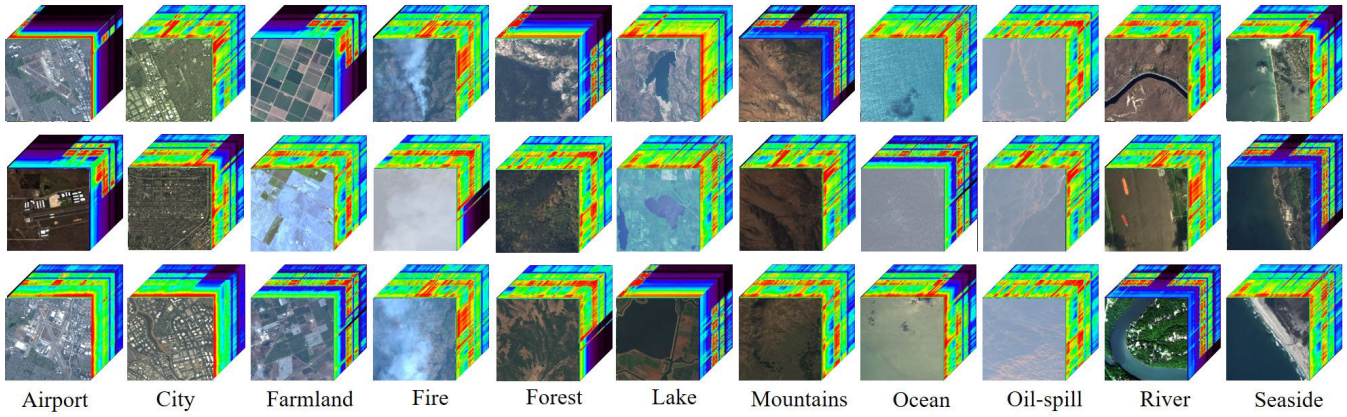


Fig. 2. Three representative samples of HRSSC dataset for each class.

Algorithm 2 SSFNet: Spectral–Spatial Fusion Network for Hyperspectral Remote Sensing Scene Classification

Input:
Input Hyperspectral image **I**
Output:
Classification score **C**
Begin
According to Eq. (1), using MSRGS to extract local hyperspectral data **M**.
According to Eq. (2), using the averaging pooling operation to extract spectral feature $\mathbf{S}_l$.
According to Eq. (3), using deep feature extraction technique to extract the local spatial feature $\mathbf{F}_l$.
According to Eq. (4), using deep feature extraction technique to extract the global spatial feature $\mathbf{F}_g$.
According to Eq. (5), using feature fusion operation followed by FC layer to calculate the class score **C**.
Return C
End

experiments to evaluate the effectiveness of the SSFNet framework and the MSRGS algorithm. Finally, the experimental results for all considered approaches are discussed.

A. HRSSC Dataset

To assess the scene classification performance of hyperspectral remote sensing images, we create a novel large-scale hyperspectral RSSC dataset based on the airborne visible/infrared imaging spectrometer (AVIRIS) hyperspectral imaging sensor, named HRSSC. This dataset consists of a total number of 1445 hyperspectral images, which are divided into 11 categories, including airport, city, farmland, fire, forest, lake, mountains, ocean, oil spill, river, and seaside. The spatial size of each hyperspectral cube is 256 × 256, and the spectral number of the hyperspectral image is 224. The spectral range is varying from 400 to 2500 nm, and the spectral resolution is 10 nm. Because of the different flight heights during data collection, their spatial resolution varies from 3 to 15 m. Fig. 2 shows several examples of each scene. It can be seen from Fig. 2 that the establishment of this HRSSC dataset

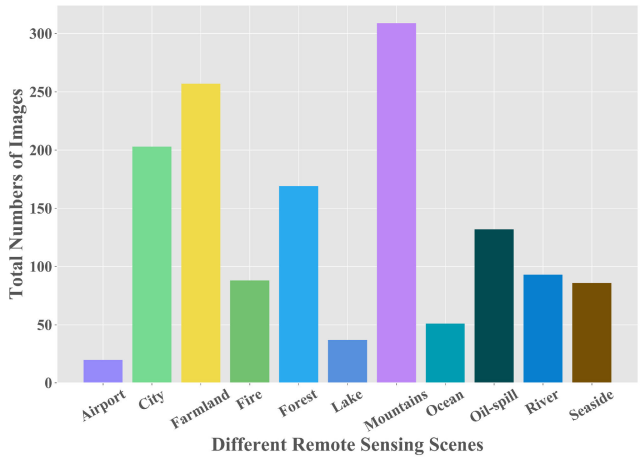


Fig. 3. Quantity distribution of each scenario in HRSSC dataset.

will help to provide the spatial distribution of land covers within an area of interest. This information can be used for urban planning, environmental monitoring, and disaster response. The number of each class in the HRSSC dataset is displayed in Fig. 3. We can clearly see that three scenes, that is, airport, lake, and ocean, include fewer hyperspectral images while city, farmland, and mountain scenes have more images. How to accurately classify different hyperspectral scenes under imbalanced remote sensing images is an interesting topic.

B. Evaluation Metrics

To quantitatively analyze the classification effectiveness of different RSSC methods, two widely used indicators, that is, overall accuracy (OA) and confusion matrix (CM), are adopted.

1) Overall Accuracy: The OA represents the ratio of the total amount of correctly identified data to the overall data. The OA provides an evaluation of the overall performance of classification methods rather than individual class.

$$\mathrm{OA} = \sum_{i=1}^{T} \frac{M_{ii}}{N} \tag{16}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:56:11 UTC from IEEE Xplore. Restrictions apply.

TABLE I
OA (%) OBTAINED BY THE PROPOSED CLASSIFICATION FRAMEWORK WITH DIFFERENT PARAMETERS T ON HRSSC DATASET

Methods	T=60%	T=70%	T=80%
GoogleNet_g	82.47	82.47	82.47
GoogleNet_{gls}	86.14	**86.52**	85.99
MIT_g	84.38	84.38	84.38
MIT_{gls}	87.14	**87.29**	87.14

TABLE II
OA (%) OBTAINED BY THE PROPOSED CLASSIFICATION FRAMEWORK WITH DIFFERENT BACKBONES AND FEATURE EXTRACTION BRANCHES ON THE HRSSC DATASET

Branch \ Backbone	ResNet18	AlexNet	GoogleNet	MIT
Global	79.40	78.02	82.47	84.38
Local	81.47	77.19	55.59	85.91
Spectral	58.12	75.36	78.10	79.25
Global+Local	80.13	79.81	85.07	86.63
Global+Spectral	81.32	79.05	82.39	74.12
Local+Spectral	81.78	68.26	76.88	86.63
Global+Local+Spectral	**82.31**	**80.25**	**86.52**	**87.29**

where T is the total number of classes. M denotes the number of correctly classified samples. N indicates the total number of samples.

2) Confusion Matrix: The CM is a comprehensive 2-D table that shows how well a classification model is performing by comparing its predictions to the actual results, which is frequently used in machine learning and classification problems. It can accurately evaluate the results of different RSSC methods and analyze the real classification situation of each category. The CM is based on the relationship between the predicted results and the real labels. The elements in the diagonal represent the correct proportion of classification, while the elements in other positions represent misclassification. The CM provides valuable insights into the classification performance for each class.

C. Experiment Setup

1) Compared Approaches: To verify the effectiveness of the SSFNet classification framework, several popular or recently proposed RSSC methods are adopted for comparison, including ResNet50 [59], self-attention-based feature fusion (SAFF) method [20], ResNet18 [18], VGG-VD-16 [29], and GCSA deep learning network (GCSANet) [23]. All methods are implemented in a personal computer with 64-GB memory CPU and 1 × 24-GB NVIDIA GeForce GTX 3090Ti GPU.

2) Parameters Setting: In the experiment, the Adam [60] is employed as the optimizer for the network training, and the cross entropy loss function is utilized to update the SSFNet model. The amount of iterations is set to 50 epochs, and the batch size of each round is 16. The Adam's initial learning rate is 1e-4 and is decreased by ten times every 20 epochs. The input size of the global stream is 256 × 256. The input data of the local branch is cropped from the original image, and then, is upsampled to 256 × 256 in the feature extraction stage. For the sake of fairness, the input size of all considered methods is 256 × 256.

D. Ablation Experiments

1) Analysis of the Influence of Hyperparameter T: In the proposed MSRGS, there is a hyperparameter T that should be determined. An experiment is conducted on the HRSSC dataset, where the training sample size is 10%. The parameter T is set to be {60%, 70%, and 80%}. Table I shows the OAs obtained by the SSFNet classification framework with different parameters T. It is obvious that when the parameter T is set to be 70%, our method can obtain the highest classification accuracy. Therefore, in the subsequent experiments, the hyperparameter T is set to be 70%.

2) Analysis of the Influence of Different Backbones: In this subsection, the influence of the SSFNet framework with different feature extractors on the scene classification effect is discussed. Several widely used feature extractors, that is, ResNet18 [61], AlexNet [62], GoogleNet [63], and MIT [58], are adopted as backbone. An experiment is conducted on the HRSSC dataset with 10% training set. Table II presents the OAs obtained by the proposed SSFNet with those backbones. It can be observed that the SSFNet framework with AlexNet shows the lowest OA. This is because the AlexNet is a relatively shallow CNN. The ResNet contains residual connections, which slightly improves the classification result. The GoogleNet yields better classification performance over ResNet since GoogleNet has a deeper architecture and multiple parallel convolutional layers. Among all feature extractors, MIT obtains the highest OA since it incorporates the self-attention mechanism to capture the global information of the input and the spatial correlation between pixels.

3) Analysis of the Influence of Each Feature Extraction Branch: Furthermore, the influence of the SSFNet framework with different branches on the classification performance is analyzed. Table II shows the experimental results obtained by the SSFNet framework with different feature extraction branches. From this table, several conclusions can be easily drawn. First, single spatial feature or spectral feature extraction fails to achieve satisfactory image classification performance. This is because the single feature cannot well represent rich spatial and spectral information in HSIs. Second, when the global and local spatial features are combined, the classification effect is improved. For example, the OA obtained by MIT with global and local features is increased by 2.25% than one with only global features. Moreover, when the global spatial and spectral features are jointly utilized, the experimental results obtained by the SSFNet framework obtain significant improvement. Third, when the global and local spatial features as well as the spectral feature are both taken into consideration, the proposed SSFNet framework yields the best scene classification accuracy. In addition, it is found that the scene classification accuracy of the SSFNet method with GoogleNet and MIT is much better than others. In the subsequent experiments, we adopt GoogleNet and MIT as feature extractors.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:56:11 UTC from IEEE Xplore. Restrictions apply.

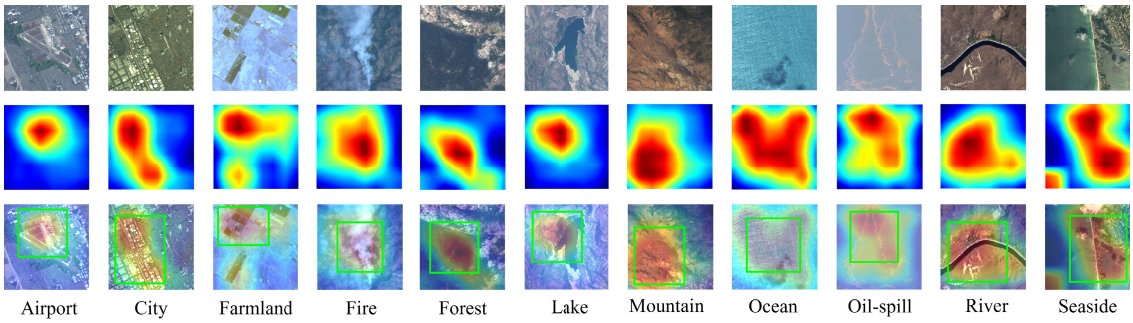


Fig. 4. Visualization result of MSRGS. The first row is the original image. The second row is the energy map. The last row is the fusion image with bounding box labels.

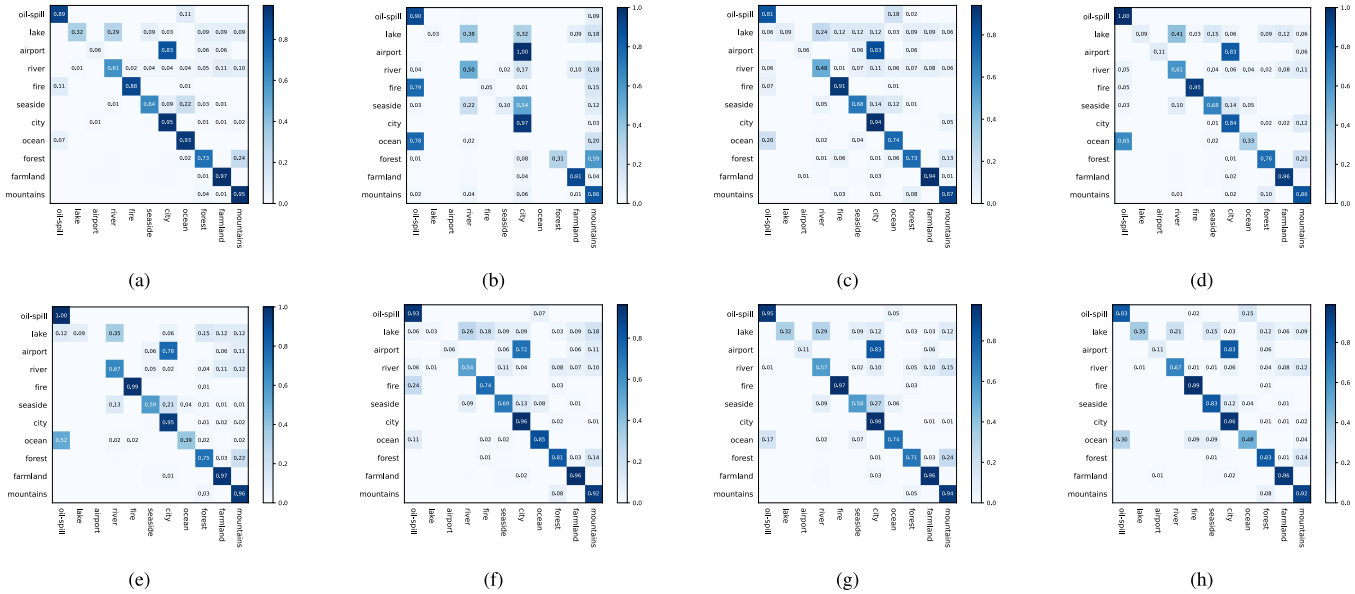


Fig. 5. CMs of different classification methods on the HRSSC dataset with 10% training set. (a) ResNet50 [59]. (b) SAFF [20]. (c) ARCNet [21]. (d) ResNet18 [18]. (e) VGG-VD-16 [29]. (f) GCSANet [23]. (g) Proposed framework with GoogleNet backbone. (h) Proposed framework with MIT backbone.

E. Visualization

Fig. 4 shows some samples of the key area location based on MSRGS. The first row is the false color composite of HSIs, which consists of multiple similar classes. The next row is the energy maps of all samples. We can see that the MSRGS method can effectively focus on important area with recognizability. For example, in the fire scene, after adding the MSRGS module, we can effectively focus on the fire point and smoke. In the lake and river scenes, the location of lakes and rivers can be accurately found, which is helpful for scene recognition. In the ocean and oil spill scenarios, they have very similar spectrum. When the MSRGS method is used, we can extract the oil spill area that is used to distinguish the oil spills and seawater. Generally, the MSRGS acts as a key role in discriminating similar scenes since it can extract the salient object region from the original image.

F. Comparison With Other Methods

Table III shows the OAs of all considered classification approaches. It is obvious that the SSFNet with MIT yields the best OA among all compared methods. Moreover, as the proportion of training samples increases, the OAs of all classification models tend to improve. The proposed method always yields higher OAs among all methods. For example, when we set the training sample ratio to 10%, the proposed method with MIT achieves a gain of 3.91% compared to the GCSANet. Moreover, the proposed method with MIT is increased by 2.25% with respect to the GCSANet, when the training sample ratio is 20%. Besides, the computational cost of different methods for each hyperspectral image in the test

TABLE III

OAs AND TESTING TIME (s) OBTAINED BY DIFFERENT CLASSIFICATION APPROACHES ON HRSSC DATASET

Methods	10% training	20% training	Time(s)
GoogleNet$_g$	82.47	86.56	0.75
GoogleNet$_{gls}$	**86.52**	**88.63**	0.77
MIT$_g$	84.38	88.29	0.83
MIT$_{gls}$	**87.29**	**89.84**	0.86
ResNet50 [59]	84.68	87.68	0.71
SAFF [20]	64.25	72.00	**0.07**
ARCNet [21]	82.47	83.11	0.70
ResNet18$_{global+local}$ [18]	80.43	82.60	0.72
VGG-VD-16 [29]	85.68	86.74	0.72
GCSANet [23]	83.38	87.59	0.75

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:56:11 UTC from IEEE Xplore. Restrictions apply.

phase is presented in Table III. Among all studied methods, the SAFF method performs the fastest in terms of testing time. This is because the SAFF method only uses a simple spectral classifier for scene classification during the testing stage. Although our method is nearly 0.8 s slower than the SAFF method, the OA obtained by our method improved by nearly 20% points with respect to the SAFF method. The main reason is that a three-branch network structure is utilized, which can fully utilize the spatial and spectral information from the original image. In general, the testing time of the SSFNet is acceptable.

In addition, to evaluate the classification effect of each category, Fig. 5 presents the CMs of all considered approaches, in which the number of training samples is 10%. As can be seen in Fig. 5, the ResNet50 method cannot well identify the lake and airport classes. The lake class is misclassified into the river class. The airport class is misclassified into the city class. This is due to the fact that these classes have very high spatial similarity. The SAFF method yields serious misclassification results for many classes, such as a lake, airport, fire, seaside, ocean, and forest. For example, the airport class is completely misclassified into city class. The ARCNet method improves the classification performance. However, the lake and airport still cannot be well identified. The ResNet18 method fails to identify the ocean, lake, and airport classes. The VGG-VD-16 method produces similar classification results. For the GCSANet method, the CM map is improved. Nevertheless, the lake and airport classes suffer from serious misclassification. In general, the SSFNet method can achieve great improvement for most of the classes. The above experiment results also further reveal the superiority of the proposed classification framework.

V. Conclusion

In this article, an SSFNet is developed for HRSSC, which involves three main steps. First, an MSRGS method is proposed to extract the local object region. Then, a three-branch network architecture is proposed to mine the global spatial, local spatial, and spectral features. Finally, the class scores obtained by performing the fully connected layer on three features are merged to generate the final classification score. More importantly, we create a hyperspectral RSSC database, which will be beneficial for the advancement of hyperspectral RSSC field. Experiments on the constructed HRSSC database illustrate that the proposed SSFNet framework can gain superior classification result compared to other scene classification approaches. Moreover, the global spatial, local spatial, and spectral features can both make a great contribution to the classification performance. However, there is a room for improving the transfer ability of the proposed model via introducing domain adaptation.

References

[1] S. Mei, Z. Han, M. Ma, F. Xu, and X. Li, "A novel center-boundary metric loss to learn discriminative features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508416.

[2] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[3] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509612.

[4] Y. Zhang, G. Jiang, Z. Cai, and Y. Zhou, "Bipartite graph-based projected clustering with local region guidance for hyperspectral imagery," *IEEE Trans. Multimedia*, vol. 26, pp. 9551–9563, 2024.

[5] Y. Zhang, S. Yan, L. Zhang, and B. Du, "Fast projected fuzzy clustering with anchor guidance for multimodal remote sensing imagery," *IEEE Trans. Image Process.*, vol. 33, pp. 4640–4653, 2024.

[6] G. Li, Z. Bai, Z. Liu, X. Zhang, and H. Ling, "Salient object detection in optical remote sensing images driven by transformer," *IEEE Trans. Image Process.*, vol. 32, pp. 5257–5269, 2023.

[7] Y. Wang et al., "Learning oriented object detection via naive geometric computing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 8, pp. 10513–10525, Aug. 2023.

[8] J. Zhang, Z. Fang, H. Sun, and Z. Wang, "Adaptive semantic-enhanced transformer for image captioning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1785–1796, Feb. 2022.

[9] W. Jiang, M. Zhu, Y. Fang, G. Shi, X. Zhao, and Y. Liu, "Visual cluster grounding for image captioning," *IEEE Trans. Image Process.*, vol. 31, pp. 3920–3934, 2022.

[10] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.

[11] S. Feng, H. Zhang, B. Xi, C. Zhao, Y. Li, and J. Chanussot, "Cross-domain few-shot learning based on decoupled knowledge distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534414.

[12] B. Qin, S. Feng, C. Zhao, B. Xi, W. Li, and R. Tao, "FDGNet: Frequency disentanglement and data geometry for domain generalization in cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Aug. 26, 2024, doi: 10.1109/TNNLS.2024.3445136.

[13] K. Xu, H. Huang, P. Deng, and Y. Li, "Deep feature aggregation framework driven by graph convolutional network for scene classification in remote sensing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5751–5765, Oct. 2022.

[14] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

[15] B. Liu, Y. Sun, A. Yu, Z. Xue, and X. Zuo, "Hyperspectral meets optical flow: Spectral flow extraction for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 5181–5196, 2023.

[16] D. Wang, B. Du, and L. Zhang, "Spectral–spatial global graph reasoning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 12924–12937, Sep. 2023.

[17] R. Minetto, M. Pamplona Segundo, and S. Sarkar, "Hydra: An ensemble of convolutional neural networks for geospatial land classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6530–6541, Sep. 2019.

[18] Q. Wang, W. Huang, Z. Xiong, and X. Li, "Looking closer at the scene: Multiscale representation learning for remote sensing image scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 4, pp. 1414–1428, Apr. 2022.

[19] K. Xu, H. Huang, and P. Deng, "Remote sensing image scene classification based on global–local dual-branch structure model," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[20] R. Cao, L. Fang, T. Lu, and N. He, "Self-Attention-Based deep feature fusion for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 43–47, Jan. 2021.

[21] Q. Wang, S. Liu, J. Chanussot, and X. Li, "Scene classification with recurrent attention of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1155–1167, Feb. 2018.

[22] X. Tang, Q. Ma, X. Zhang, F. Liu, J. Ma, and L. Jiao, "Attention consistent network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2030–2045, 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:56:11 UTC from IEEE Xplore. Restrictions apply.

[23] W. Chen, S. Ouyang, W. Tong, X. Li, X. Zheng, and L. Wang, "GCSANet: A global context spatial attention deep learning network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1150–1162, 2022.

[24] N. He, L. Fang, S. Li, J. Plaza, and A. Plaza, "Skip-connected covariance network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1461–1474, May 2020.

[25] J. Fang, Y. Yuan, X. Lu, and Y. Feng, "Robust space–frequency joint representation for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7492–7502, Oct. 2019.

[26] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, Nov. 2010, pp. 270–279.

[27] G. Sheng, W. Yang, T. Xu, and H. Sun, "High-resolution satellite scene classification using a sparse coding based multiple feature combination," *Int. J. Remote Sens.*, vol. 33, no. 8, pp. 2395–2412, Apr. 2012.

[28] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.

[29] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[30] P. Duan, S. Hu, X. Kang, and S. Li, "Shadow removal of hyperspectral remote sensing images with multiexposure fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537211.

[31] X. Kang, Y. Zhu, P. Duan, and S. Li, "Two-dimensional spectral representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502809.

[32] T.-H. Chan, K. Jia, S. Gao, J. Lu, Z. Zeng, and Y. Ma, "PCANet: A simple deep learning baseline for image classification?" *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 5017–5032, Dec. 2015.

[33] P. Duan, X. Kang, P. Ghamisi, and S. Li, "Hyperspectral remote sensing benchmark database for oil spill detection with an isolation forest-guided unsupervised detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509711.

[34] P. Duan, T. Shan, X. Kang, and S. Li, "Spectral super-resolution in frequency domain," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Oct. 29, 2024, doi: 10.1109/TNNLS.2024.3481060.

[35] Z. Li, W. Liu, Z. Xie, X. Kang, P. Duan, and S. Li, "FAA-det: Feature augmentation and alignment for anchor-free oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5539411.

[36] L. Huang, C. Chen, W. Li, and Q. Du, "Remote sensing image scene classification using multi-scale completed local binary patterns and Fisher vectors," *Remote Sens.*, vol. 8, no. 6, p. 483, Jun. 2016.

[37] Y. Li, L. Fan, and W. Xie, "TGSIFT: Robust SIFT descriptor based on tensor gradient for hyperspectral images," *Chin. J. Electron.*, vol. 29, no. 5, pp. 916–925, Sep. 2020.

[38] B. Luo, S. Jiang, and L. Zhang, "Indexing of remote sensing images with different resolutions by multiple features," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 4, pp. 1899–1912, Aug. 2013.

[39] J. Zou, W. Li, C. Chen, and Q. Du, "Scene classification using local and global features with collaborative representation fusion," *Inf. Sci.*, vol. 348, pp. 209–226, Jun. 2016.

[40] B. Zhao, Y. Zhong, and L. Zhang, "A spectral–structural bag-of-features scene classifier for very high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 116, pp. 73–85, Jun. 2016.

[41] X. Zheng, X. Sun, K. Fu, and H. Wang, "Automatic annotation of satellite images via multifeature joint sparse coding with spatial relation constraint," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 4, pp. 652–656, Jul. 2013.

[42] F. Zhang, B. Du, and L. Zhang, "Saliency-guided unsupervised feature learning for scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2175–2184, Apr. 2015.

[43] R. Bahmanyar, S. Cui, and M. Datcu, "A comparative study of bag-of-words and bag-of-topics models of EO image patches," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 6, pp. 1357–1361, Jun. 2015.

[44] W. Huang, Q. Wang, and X. Li, "Feature sparsity in convolutional neural networks for scene classification of remote sensing image," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 3017–3020.

[45] J. Li et al., "Deep discriminative representation learning with attention map for scene classification," *Remote Sens.*, vol. 12, no. 9, p. 1366, Apr. 2020.

[46] Y. Yu and F. Liu, "A two-stream deep fusion framework for high-resolution aerial scene classification," *Comput. Intell. Neurosci.*, vol. 2018, pp. 1–13, Aug. 2018.

[47] C. Zhao et al., "Hyperspectral image classification with multi-attention transformer and adaptive superpixel segmentation-based active learning," *IEEE Trans. Image Process.*, vol. 32, pp. 3606–3621, 2023.

[48] Q. Zhao, S. Lyu, Y. Li, Y. Ma, and L. Chen, "MGML: Multigranularity multilevel feature ensemble network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 5, pp. 2308–2322, May 2023.

[49] Y. Zhang, X. Zheng, Y. Yuan, and X. Lu, "Attribute-cooperated convolutional neural network for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8358–8371, Dec. 2020.

[50] H. Zhu, W. Ma, L. Li, L. Jiao, S. Yang, and B. Hou, "A dual–branch attention fusion deep network for multiresolution remote-sensing image classification," *Inf. Fusion*, vol. 58, pp. 116–131, Jun. 2020.

[51] R. Zhu, L. Yan, N. Mo, and Y. Liu, "Attention-based deep feature fusion for the scene classification of high-resolution remote sensing images," *Remote Sens.*, vol. 11, no. 17, p. 1996, Aug. 2019.

[52] G. Zhang et al., "A multiscale attention network for remote sensing scene images classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9530–9545, 2021.

[53] X. Zhang, X. Fan, G. Wang, P. Chen, X. Tang, and L. Jiao, "MFGNet: Multibranch feature generation networks for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5609613.

[54] Y. Xiong, K. Xu, Y. Dou, Y. Zhao, and Z. Gao, "WRMatch: Improving FixMatch with weighted nuclear-norm regularization for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5612214.

[55] W. Li, X. Wang, G. Li, B. Geng, and P. K. Varshney, "A copula-guided in-model interpretable neural network for change detection in heterogeneous remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4700817.

[56] X. Li, D. Shi, X. Diao, and H. Xu, "SCL-MLNet: Boosting few-shot remote sensing scene classification via self-supervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5801112.

[57] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.

[58] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

[59] R. M. Anwer, F. S. Khan, J. van de Weijer, M. Molinier, and J. Laaksonen, "Binary patterns encoded convolutional neural networks for texture recognition and remote sensing scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 138, pp. 74–85, Apr. 2018.

[60] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[61] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[62] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[63] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Apr. 2015, pp. 1–9.



Zhen Li received the M.S. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2011. She is currently pursuing the Ph.D. degree with the College of Computer Science and Technology (CCST), Zhejiang University, Hangzhou, China.

She is employed by the Institute of Remote Sensing Satellite, CAST, Beijing, China. Her research interests include the design of intelligent remote sensing satellites, especially intelligent applications based on hyperspectral sensors.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:56:11 UTC from IEEE Xplore. Restrictions apply.



Puhong Duan (Member, IEEE) received the B.S. degree from Suzhou University, Suzhou, China, in 2014, and the Ph.D. degree from Hunan University, Changsha, China, in 2021, respectively.

From October 2019 to December 2019, he was a Visiting Ph.D. Student with the Faculty of Electrical Engineering and Computer Science, Technische Universitat Berlin, Berlin, Germany. From January 2020 to October 2020, he was a Visiting Ph.D. Student with the Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, Freiberg, Germany. He is currently an Associate Professor with the School of Robotics, Hunan University. His research interests include image reconstruction, multimodal image fusion, and classification.

Dr. Duan was a finalist for the Best Student Paper Award at the International Geoscience and Remote Sensing Symposium (IGARSS) 2020 and selected as the Best Reviewer for IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2021.



Jialin Zheng received the M.S. degree from the College of Electrical and Information Engineering, Hunan University, Changsha, China, in 2024.

He is currently an Assistant Engineer with Pinghu Laboratory, Pinghu, Zhejiang, China. His research interests include remote sensing image classification and object detection.



Zhuojun Xie received the B.Sc. degree from Lanzhou University of Technology, Lanzhou, China, in 2019, and the Ph.D. degree from Hunan University, Changsha, China, in 2024.

His research interests include cross-scene hyperspectral image classification and open-set recognition.



Xudong Kang (Senior Member, IEEE) received the B.Sc. degree from Northeast University, Shenyang, China, in 2007, and the Ph.D. degree from Hunan University, Changsha, China, in 2015.

In 2015, he joined the College of Electrical Engineering, Hunan University. His research interests include hyperspectral feature extraction, image classification, image fusion, and anomaly detection.

Dr. Kang received the National Natural Science Award of China (second class and rank as third) and second prize in the Student Paper Competition in IGARSS 2014. He was selected as the Best Reviewer for IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He was an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2018 to 2019. He serves as an Associate Editor for IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and IEEE JOURNAL ON MINIATURIZATION FOR AIR AND SPACE SYSTEMS.



Jianwei Yin (Member, IEEE) received the Ph.D. degree in computer science from Zhejiang University (ZJU), Ningbo, China, in 2001.

He was a Visiting Scholar with Georgia Institute of Technology, Atlanta, GA, USA. He is currently a Full Professor with the College of Computer Science, ZJU. He has published more than 100 articles in top international journals and conferences. His research interests include quantum computing, service computing, and business process management.

Dr. Yin is an Associate Editor of IEEE TRANSACTIONS ON SERVICES COMPUTING.



Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University. He is currently a Full Professor with the College of Electrical and Information Engineering, Hunan University. He has authored or co-authored over 200 refereed articles. His research interests include image processing, pattern recognition, and artificial intelligence.

Dr. Li is a member of the Editorial Board of *Information Fusion* and *Sensing and Imaging*. He received two Second-Grade State Scientific and Technological Progress Awards of China in 2004 and 2006. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:56:11 UTC from IEEE Xplore. Restrictions apply.